

Junghye Kim

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github.com/junghye01

EDUCATION

M.S. in Artificial Intelligence, Korea University, Seoul, Korea (2025.09 - 2027.08)

Advisor : Sungwoong Kim, Sungbin Lim

Research Focus : Multimodal Learning, Video Understanding, and Visual Reasoning

B.S. in Dongguk University, Seoul, Korea (2020.02-2024.02)

Department of Information & Communication Engineering

Double major in DataScience software

Cumulative GPA: 4.26/4.5

Honors : Summa Cum Laude

RESEARCH INTERESTS

Multimodal Reasoning, Video Understanding, Human-Object Interaction, Temporal and Causal Reasoning

RESEARCH EXPERIENCE

Graduate Researcher, AGILAB, Korea University (2025.09-Present)

Developed structured reasoning and spatio-temporal grounding methods for video reward modeling, resulting in a co-first-author EMNLP 2026 Main Conference paper.

AGILAB Intern, Korea University (2025.04-2025.08)

Conducted research on video reward modeling for capturing fine-grained temporal dynamics and efficient inference techniques for large language models.

Researcher, Seoul Asan Medical Center (2024.08-2025.04)

Conducted CT-based cancer diagnosis research by developing a 3D benign lymph node segmentation model and a weakly supervised liver tumor segmentation model using synthetic tumor generation.

Research Intern, MEDAI (2023.11-2023.12)

Conducted research on CT reconstruction and a self-supervised CT image interpolation framework using image masked modeling.

Researcher, Wbridge Engineering Research Program (2023.04-2023.10)

Developed an adaptive prompt-matching technique that dynamically applies prompts based on question types; resulted in a publication in KSC 2023.

Data Analysis & Machine Intelligence LAB Intern, Dongguk University (2023.01-2023.10)

Conducted machine learning and computer vision research on hierarchical network attack classification and long-tailed multi-label image classification.

Research Intern, NIA AI Data Construction Project (2022.08-2022.12)

Conducted research on 3D human action analysis by developing the DGU-HAO benchmark dataset and systematically evaluating its effectiveness using skeleton-based action recognition models; resulted in a publication in IEEE Access.

PUBLICATIONS

Plan-and-Verify Video Reward Reasoning with Spatio-Temporal Scene Graph Grounding

Hyomin Kim*, Junghye Kim*, Joanie Hayoun Chung, Yoonjin Oh, Kyungjae Lee, Sungbin Lim, Sungwoong Kim.
EMNLP 2026 — Main Conference (* Equal contribution)

DGU-HAO : A Dataset with Daily Life Objects for Comprehensive 3D Human Action Analysis

Jiho Park, Junghye Kim, Yujung Gil, Dongho Kim
IEEE ACCESS, 2024

APMT : Adaptive Prompt-Matching Technique to enhance your answers
Yunji Kim, **Junghye Kim***, Yeosol Heo*, Jihie Kim (* : Equal Contribution)
Korea Software Congress, 2023

PATENTS

Apparatus and Method of Classifying and Detecting Real-Time Abnormal Behavior Based on Artificial Intelligence
(No. 10-0004987(2024)) (Submitted) Sangsoo Lim, **Junghye Kim**, Junho Choi, Seunghyeon Jung

CERTIFICATIONS

Advanced Data Analytics Semi-Professional(ADsP) Certification FEB 2022

TEACHING EXPERIENCE

‘Machine Learning and Data Science’ - Teaching Assistant SEP 2023- DEC 2023
- Led practical sessions applying machine learning algorithms to real-world datasets

Wbridge, Korean Women in Science and Technology Foundation - Mentor APRIL 2023- OCT 2023
- Mentored high school students on open-source project development and collaboration using Git and GitHub
- Guided students in research team participation and study engagement

‘Open source Software Practice’ - Teaching Assistant MAR 2023- JUNE 2023
- Led practical sessions on Docker-based deployment of Flask web applications
- Mentored 50 students through Q&A and interactive discussions

HONORS AND AWARDS

2024 Dongguk University Summa cum laude
2022 Dongguk University Academic Excellence Scholarship
2021 Top Student Scholarship by Course (‘General Physics and Experiment 2’)

LANGUAGE PROFICIENCY

Fluent in **English(TOEIC 905)** and Native in **Korean**

TECHNICAL SKILLS

Languages: Python, C, Javascript
Tools : Python, C, JavaScript, PyTorch, Hugging Face Transformers, OpenCV, NumPy, Git, GitHub, Docker, AWS, Jenkins